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1.0 Abstract

This report presents a comprehensive logical design for a commuter transport optimization and fare-tracking system tailored to address urban public transportation challenges in Sierra Leone. Prepared as an individual assignment for PROG101: Principles of Programming Logic and Design, this project utilizes computational thinking, problem decomposition, modular design, pseudocode, and algorithmic flowcharting without writing physical source code. The proposed system—designated as the Commuter Transit Management System—targets operational inefficiencies in commercial minibuses (poda-podas) and taxis, such as unregulated fare fluctuations, vehicle overcrowding, and lack of transparent passenger tracking. Furthermore, the design incorporates Open Source Software (OSS) principles and Digital Public Goods (DPGs) frameworks to ensure long-term community sustainability, full alignment with United Nations Sustainable Development Goal 11 (Sustainable Cities and Communities), and transparent project management through GitHub version control.

2.0 Introduction & Rationale

2.1 Problem Description & Background

Public transportation in Sierra Leone, particularly across major urban routes in Freetown, relies heavily on informal commercial transport networks known locally as poda-podas and taxis. Commuters frequently experience severe operational bottlenecks, including arbitrary and unregulated fare increases during peak hours, vehicle overcrowding, and lack of predictability in route scheduling.

2.2 Community Impact & Beneficiaries

These transportation inefficiencies cause daily economic strain and safety risks for everyday commuters, students, and workers. The primary beneficiaries of the proposed logical system include everyday commuters, transport union operators, and regulatory authorities who require transparent tracking of passenger manifests and standardized fare structures.

2.3 Proposed Solution & SDG Alignment

Acting as a Junior System Analyst, this project designs a robust, logic-driven transport management system that processes route distances, passenger loads, and time-of-day variables. The project directly supports **United Nations Sustainable Development Goal 11 (SDG 11): Sustainable Cities and Communities** by fostering accessible, safe, and efficient public transit systems.

3.0 Computational Thinking & Algorithm Design

3.1 Problem Decomposition & Abstraction

Using computational thinking, the transport challenge is broken down into modular sub-tasks: Route Selection, Passenger Load Verification, Dynamic Fare Calculation, and Ticket Generation. Abstraction is applied by focusing exclusively on logical decision rules rather than hardware implementation details.

3.2 Step-by-Step Algorithm

1. Start the system and initialize route and capacity variables.
2. Input RouteID, Maximum Capacity, and current Passenger Count.
3. Check if PassengerCount >= MaxCapacity. If true, output "Vehicle Full: Boarding Closed" and stop.
4. If space is available, input Distance Zone and Peak Hour status.
5. Call the CalculateFare function to determine ticket pricing based on zone and surcharges.
6. Increment the passenger count by 1 and issue the ticket summary.
7. Repeat boarding process until capacity is reached or operator halts.
8. Stop.

4.0 System Logical Design & Pseudocode

4.1 IPO Model & Decision Structures

- **Inputs:** Route ID, Distance Zone, Passenger Count, Max Capacity, Peak Hour flag.
- **Processing:** Validation, relational checks, and nested conditional calculations.
- **Outputs:** Ticket pricing, vehicle status alerts, and trip logs.

4.2 Pseudocode Implementation

```
Plaintext
Module MainProgram()
    Declarations
        String routeID, peakHour, boardMore = "YES"
        Integer distanceZone, passengerCount, maxCapacity
        Real finalFare

    Output "Enter Route ID, Max Capacity, Current Count: "
    Input routeID, maxCapacity, passengerCount

    While boardMore == "YES" AND passengerCount < maxCapacity
        Output "Enter Zone (1:Short, 2:Med, 3:Long) & Peak (YES/NO): "
        Input distanceZone, peakHour

        finalFare = CalculateFare(distanceZone, peakHour)
        passengerCount = passengerCount + 1

        Output "Boarded Successfully. Fare: Le ", finalFare
        Output "Current Count: ", passengerCount, " / ", maxCapacity

        If passengerCount >= maxCapacity Then
            Output "WARNING: Vehicle Full. Boarding Closed."
        Else
            Output "Board another passenger? (YES/NO): "
            Input boardMore
        End If
    End While
End Module

Function Real CalculateFare(Integer zone, String isPeak)
    Real fare = (zone == 1) ? 5.00 : (zone == 2) ? 10.00 : 15.00
    If isPeak == "YES" Then fare = fare * 1.25
    Return fare
End Function
```

5.0 Flowcharts & Expected Outputs

5.1 Flowchart Logic Structure

- **Start (Oval):** Initialize variables.
- **Input (Parallelogram):** Enter RouteID, MaxCapacity, PassengerCount.
- **Decision (Diamond):** Is PassengerCount < MaxCapacity?
    - *No:* Output "Vehicle Full" → End.
    - *Yes:* Input Zone & Peak Status → Calculate Fare → Increment Count → Output Ticket.

5.2 Functional Modules & Sample Outputs

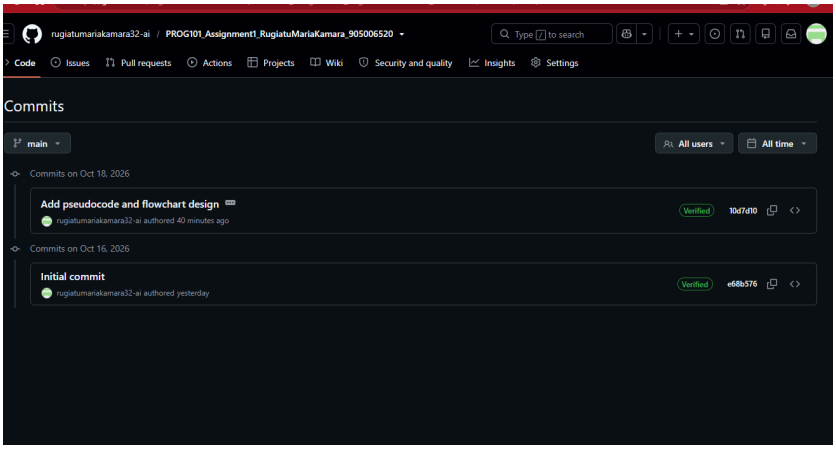
- **Scenario Input:** Route ID: "FRT-04", Zone: 2 (Medium), Peak: "YES", Initial Passengers: 13, Max Capacity: 14.
- **Processed Output:** Base fare Le 10.00 adjusted by 1.25 peak multiplier = **Le 12.50**. Passenger count updates to 14.
- **System Message:** Boarded Successfully. Fare: Le 12.50 followed by WARNING: Vehicle Full. Boarding Closed.

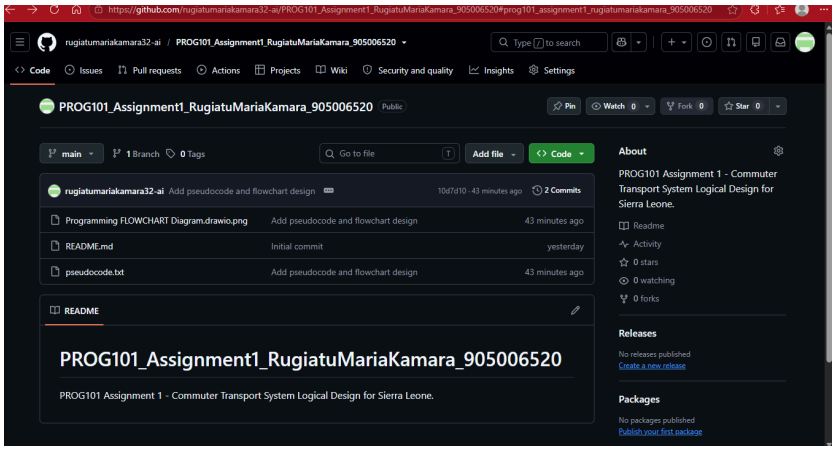
6.0 Open Source, Digital Public Goods & GitHub Evidence

6.1 Open Source & DPG Justification

Deploying this system as **Open Source Software (OSS)** eliminates expensive proprietary licensing fees for local transport unions in Sierra Leone, allows customization for regional route codes, and builds local developer capacity. Registering the architecture as a **Digital Public Good (DPG)** ensures transparency, open access, and public sector benefit.

6.2 GitHub Repository & Version Control Evidence

- **Repository Name:PROG101_Assignment1_RugiatuMariaKamara_905006520**
- **Repository URL:** rugiatumariakamara32-ai/PROG101_Assignment1_RugiatuMariaKamara_905006520: PROG101 Assignment 1 - Commuter Transport System Logical Design for Sierra Leone.
- **Commit History Summary:**
  - Initial project setup and README.md creation
  - Add pseudocode and flowchart design
  - Add final project report PDF with GitHub evidence
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7.0 Conclusion

In summary, this assignment successfully applied computational thinking, structured algorithms, pseudocode, and modular design to address an urban transportation challenge in Sierra Leone. By establishing a logical framework for fare calculation and passenger capacity management, the project demonstrates how systematic problem analysis precedes effective software development. Furthermore, integrating Open Source principles, Digital Public Goods criteria, and GitHub version control ensures that the solution is sustainable, scalable, and socially impactful.

8.0 References

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